

THONG PHAPHUM, Thailand

WMO: 484210

Lat: 14.742N

Lon: 98.636E

Elev: 326

StdP: 14.52

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	59.1	61.9	53.0	60.6	72.4	55.9	67.3	74.3	10.1	85.3	8.0	85.9	0.1	N/A	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	22.3	101.4	78.2	99.5	77.3	97.7	76.9	81.8	93.2	81.0	91.5	80.6	90.5	2.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	153.3	86.0	78.7	150.6	85.2	78.2	148.1	84.5	45.8	93.4	45.0	92.1	44.4	90.5	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	7.0	5.8	DB	54.5	104.1	3.9	2.1	51.7	105.6	49.4	106.8	47.2	107.9	44.4	109.4	(4)
(5)			WB	53.3	84.5	3.8	2.3	50.5	86.1	48.3	87.4	46.1	88.7	43.4	90.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.0	78.4	81.9	85.1	87.1	84.8	82.5	81.2	81.1	82.0	82.2	80.2	77.6	(6)
(7)		DBStd	3.88	3.69	3.25	2.72	2.81	3.49	2.39	2.09	2.11	1.99	2.04	2.92	4.02	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11682	880	892	1088	1113	1080	976	968	964	961	998	907	855	(10)
(11)		CDD65	6207	415	472	623	663	615	526	503	499	511	533	457	390	(11)
(12)		CDH74	63512	3990	5345	7738	8432	6856	5209	4441	4294	4580	4956	4162	3509	(12)
(13)		CDH80	28350	1925	2927	4476	4796	3123	1783	1275	1190	1522	1940	1829	1564	(13)
(14)	Wind	WSAvg	0.5	0.5	0.8	1.0	0.8	0.4	0.3	0.3	0.4	0.4	0.3	0.5	0.6	(14)
(15)	Precipitation	PrecAvg	90.4	0.3	0.5	2.2	3.9	10.4	14.4	18.9	18.7	13.0	7.0	0.9	0.2	(15)
(16)		PrecMax	110.3	1.3	3.0	7.6	14.2	18.0	26.9	35.2	31.2	22.2	13.0	3.6	1.7	(16)
(17)		PrecMin	60.7	0.0	0.0	0.0	0.4	3.6	9.5	7.0	6.7	6.9	1.0	0.0	0.0	(17)
(18)		PrecStd	12.7	0.4	0.7	1.8	2.7	3.6	3.7	6.2	4.8	3.8	2.8	0.9	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.8	99.4	102.4	104.4	103.2	96.2	93.2	92.9	93.5	94.2	95.0	94.4	(19)
(20)			MCWB	75.0	74.9	76.9	79.2	80.1	80.2	79.9	79.7	80.2	79.8	78.9	76.9	(20)
(21)		2%	DB	93.9	97.6	100.5	102.0	99.5	93.0	90.6	90.5	91.7	92.5	93.1	92.6	(21)
(22)			MCWB	74.7	74.5	76.2	79.0	79.8	79.9	79.5	79.3	79.7	79.5	78.1	76.0	(22)
(23)		5%	DB	92.2	96.0	98.9	100.0	96.0	90.5	88.4	88.4	90.0	91.2	91.3	90.7	(23)
(24)			MCWB	74.1	74.2	75.6	78.6	79.8	79.6	79.1	79.0	79.5	79.1	77.1	74.9	(24)
(25)		10%	DB	90.2	94.1	97.1	97.8	92.9	88.2	86.3	86.1	87.8	89.4	89.4	88.6	(25)
(26)			MCWB	73.4	73.8	75.3	78.5	79.7	79.3	78.7	78.6	79.1	78.8	76.4	73.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.6	79.1	81.0	83.3	82.9	81.8	81.0	81.0	81.6	81.7	81.0	79.8	(27)
(28)			MCDWB	90.2	93.3	97.2	99.3	95.5	90.5	88.9	88.9	90.1	90.9	91.2	90.1	(28)
(29)		2%	WB	77.2	77.9	79.4	81.5	81.6	80.9	80.2	80.1	80.6	80.8	79.8	78.0	(29)
(30)			MCDWB	88.6	91.5	94.7	96.6	93.2	89.1	87.5	87.5	89.1	89.9	89.6	88.7	(30)
(31)		5%	WB	76.1	76.9	78.4	80.5	80.7	80.3	79.7	79.5	80.0	80.1	78.9	76.9	(31)
(32)			MCDWB	87.3	90.3	92.9	94.9	91.5	88.0	86.4	86.2	87.8	88.7	88.3	87.2	(32)
(33)		10%	WB	75.0	75.9	77.5	79.7	80.0	79.7	79.0	78.9	79.3	79.4	77.9	75.8	(33)
(34)			MCDWB	86.0	88.8	91.2	93.2	89.8	86.7	85.2	84.9	86.4	87.2	86.2	85.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	25.5	26.7	25.3	22.3	15.7	12.1	11.0	10.9	12.9	15.6	19.6	23.2	(35)
(36)			MCDBR	26.5	27.9	27.9	25.5	21.4	15.9	14.4	14.5	15.9	17.9	21.2	24.0	(36)
(37)			MCWBR	9.5	8.4	8.1	7.1	5.5	5.1	5.2	5.0	5.5	6.4	7.8	9.0	(37)
(38)		5% WB	MCDBR	22.9	24.0	24.2	22.5	17.7	13.5	12.6	12.6	14.4	16.1	18.8	21.4	(38)
(39)	MCWBR		8.7	7.9	7.6	7.3	5.6	5.1	5.3	5.0	5.6	6.2	7.4	8.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.499	0.587	0.672	0.577	0.484	0.448	0.443	0.443	0.448	0.537	0.482	0.477	(40)	
(41)		taud	2.060	1.858	1.687	1.962	2.269	2.386	2.398	2.386	2.361	2.058	2.190	2.147	(41)	
(42)		Ebn at Noon	252	235	219	241	261	267	269	272	272	245	255	254	(42)	
(43)		Edh at Noon	51	65	80	60	44	38	38	39	40	53	45	46	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1536	1695	1785	1883	1515	1295	1177	1181	1319	1458	1531	1503	(44)	
(45)		RadStd	72	74	124	114	169	155	149	127	122	94	89	61	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.59	N/A	N/A	N/A	N/A

Nomenclature: See separate page